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An integrated TOPSIS framework with Full-Range Weight Sensitivity Analysis for robust decision analysis

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ABSTRACT

Multi-criteria decision-making (MCDM) is widely used in engineering to assist in the selection of the best alternative according to various criteria. Many MCDM methods rely on fixed weight assignments, limiting their ability to reflect uncertainties or variations in decision-maker preferences. This work shows a novel approach that integrates the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) with conventional one-at-a-time Weight Sensitivity Analysis (WSA) and, for the first time, introduces the Full-Range Weight Sensitivity Analysis (FRWSA). FRWSA enumerates all admissible weight vectors on a discretized simplex and summarizes robustness as dominance frequency, the share of the simplex for which an alternative ranks first. A wastewater treatment plant (WWTP) case study evaluating four different configurations illustrates the approach. Using FRWSA with step $h = 0.05$ (10 million weight combinations), configuration A2/O-D dominates 82.80% of the weight simplex, with UCT at 12.17%, A2/O-S at 4.94%, and BARD at 0.09%. Comparing with Monte Carlo sampling, FRWSA provides a deterministic, variance-free baseline: MC–Dirichlet with $\alpha = 0.5$ is nearly identical (JSD $\approx$ 0.001 bits), $\alpha = 1.0$ is close but distinct (JSD $\approx$ 0.007-0.008 bits), and plain MC remains farther (JSD $\approx$ 0.043 bits) over $10^4 - 10^7$ draws. The framework improves transparency via global coverage and boundary diagnostics and is method-agnostic (replicated with VIKOR in the SI). A reusable MATLAB implementation is provided to facilitate adoption. This integrated analysis supports robust and transparent engineering decisions wherever weight uncertainty matters.

1. Introduction

The increasing complexity of engineering systems challenges decision-makers to select the most suitable alternative for various applications. Engineering solutions are essential for shaping the modern world, from designing resilient infrastructure to creating innovative products and optimizing industrial processes [1]. Traditionally, the evaluation of engineering alternatives often relied on singular performance metrics or cost-effectiveness considerations. The multifaceted nature of engineering demands a holistic approach that considers multiple factors such as performance, cost-effectiveness, environmental impact, safety, system reliability, energy efficiency or adherence to regulatory standards, among others [1–5]. In this context, multi-criterion decision making (MCDM) techniques have emerged as essential tools, offering structured frameworks for informed and transparent decision-making [6]. A recent review also covers the field evolution [7] surveying three decades of MCDM sorting methods covering full-aggregation, outranking, goal/reference-level and non-classical approaches, and outline open challenges that frame the present study.

Among MCDM techniques, the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) is one of the most widely adopted due to its conceptual simplicity and effectiveness. TOPSIS, introduced by Hwang and Yoon [8] and further improved by other authors [9–12], is a robust methodology that provides a structured framework for systematically comparing and evaluating various alternatives based on multiple criteria. Thanks to its usefulness, it has been applied to many different fields [13–15]. TOPSIS evaluates alternatives based on their geometric proximity to an ideal solution (best performance) and a negative-ideal solution (worst performance), enabling a straightforward ranking of options.

In many TOPSIS applications, criteria weights are commonly assigned based on expert judgment or literature values [16–18]. However, TOPSIS is sensitive to the weights assigned to criteria, which reflect the relative importance of each evaluation factor. Incorrect or arbitrary weight assignments can significantly influence the outcome, compromising the reliability of the decision process [10,19]. To address this issue, Weight Sensitivity Analysis (WSA) has been

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applied to examine how individual weight variations affect ranking stability [20]. Yet, traditional WSA techniques typically vary one criterion at a time, lacking the ability to capture interactions across the entire weight space. Other authors have emphasized the role of decision-maker preferences by developing flexible TOPSIS-based interfaces where users can interactively adjust weights to simulate different ranking scenarios [21]; however, such approaches rely on manual input and do not systematically explore the full weight space, limiting their capacity to comprehensively assess ranking robustness under uncertain or variable preferences. More recently, Susmaga et al. [22] proposed a visual analysis framework to explain how different weight vectors and aggregation rules impact rankings in TOPSIS, further underscoring the importance of weight-related robustness in MCDM. However, such visual approaches improve explainability without enumerating neither the entire weight simplex nor quantifying coverage (e.g., the share of admissible weights for which an alternative dominates), and therefore do not yield variance-free robustness measures. In parallel, simulation-based sensitivity remains prevalent, but conclusions depend on the sampling design and parameterization (e.g., the assumed distribution over weights) and exhibit sampling variance even at large sample sizes [20]. Beyond manual or fixed weights, recent studies derive weights from data. Examples include an NWPHF-TODIM model for solid-waste selection using maximizing-deviation weighting [23]; a circular q-rung orthopair fuzzy framework for purification strategy selection with entropy- and Schweizer-Sklar-based aggregation for unknown weights [24]; a q-ROF 2-tuple VIKOR+QUALIFLEX scheme that estimates weights from partial information [25]; and a cubic q-ROF TODIM approach with Minkowski-type distances and entropy-based weighting [26]. In this study, we propose the novel Full-Range Weight Sensitivity Analysis (FRWSA), which does not select a single weight vector but deterministically enumerates a discretized weight simplex to produce variance-free dominance frequencies and reveal decision-boundary structure, complementing these other methods.

While the value of WSA within TOPSIS is recognized, there is a notable gap in the systematic exploration of the insights that can be drawn from the TOPSIS framework. The precise and systematic determination of the optimal criterion weight remains a critical issue that warrants closer scrutiny. Recent works have attempted to improve robustness assessment in MCDM. Więckowski et al. [27] evaluated the robustness of renewable energy sources by varying the criteria for weighting the scenarios, identifying the relevance of the criteria and the stability of the ranking. Yang and Chen [28] also implemented different sensitivity strategies and the Critical Method to determine the values of the weights in a systematically way to show which criteria have a higher or lower impact over the result. On the other hand, other studies have focused on sensitivity analyses when dealing with uncertain data by varying the criteria values instead of the weights [29] to evaluate their influence on the final ranking. Nonetheless, these methods often rely on sampling or partial explorations of the weight domain, which may overlook regions where alternative rankings shift.

FRWSA is presented here as a structured, exhaustive method that systematically explores all feasible combinations of weight vectors under the constraint that their sum equals one. Unlike Monte Carlo (MC) simulations or heuristic techniques, FRWSA evaluates all feasible weight vectors on a discretized simplex, offering a complete characterization of how rankings respond to weight variations. This allows us to identify not just whether an alternative is optimal under a certain set of weights, but also how robust that optimality is across the entire space.

To demonstrate the method's capabilities, we applied FRWSA to a wastewater treatment plant (WWTP) configuration selection using a real case study with four process alternative configurations. Three approaches are evaluated and compared: (i) TOPSIS with fixed weights, (ii) conventional WSA, and (iii) the proposed FRWSA. The results show that the FRWSA method provides a much deeper understanding of ranking stability and outperforms traditional WSA methods in transparency, coverage, robustness and interpretability.

Although applied here to environmental engineering, FRWSA is developed as a general-purpose method that can be extended to any decision-making problem involving multiple criteria. It contributes both methodologically and practically to the evolving landscape of MCDM tools. TOPSIS is adopted owing to its geometric transparency, ubiquity in engineering practice, compatibility with weight-focused sensitivity and as a popular MCDM method in the literature (see Section S1 in the supplementary information). Importantly, the FRWSA engine is method-agnostic: the same exhaustive weight-space scan can apply atop VIKOR, PROMETHEE, MARCOS, or WASPAS. Existing sensitivity practices are often (i) one-at-a-time (WSA), (ii) sample-based with variance and parameter dependence (MC), or (iii) visual but non-exhaustive. FRWSA contributes a deterministic, full-grid mapping of the weight simplex with robustness measured as dominance frequency and interpretable per-criterion strips. Novelty lies in the exhaustive grid exploration, the dominance-frequency robustness metric, and diagnostics that expose decision boundaries.

2. Materials and methods

2.1. TOPSIS method

TOPSIS was implemented in MATLAB® 2023b following the formulations of Li et al. [10]. Annotated source scripts for problems up to 10 criteria are provided in the supplementary information (*fun_TOPSIS.m*), and a compact mathematical formulation is included in Section S2 of the supplementary information. The detailed procedure consists of six steps.

Step 1 – Definition of criteria, weights, and MaxMin vector. The user needs to define the criteria and the associated weights in three different inputs: the decision matrix (C), the weighting vector (w) and the *MaxMin* vector (*MM*).

The decision matrix ($C \in \mathbb{R}^{m\times n}$) for m alternatives and n criteria is given in Eq. (1).

$$Decision\ matrix\ (C) = \begin{bmatrix} c_{11} & c_{12} & c_{1j} & \cdots & c_{1n} \\ c_{21} & c_{22} & c_{2j} & \cdots & c_{2n} \\ c_{i1} & c_{i2} & c_{ij} & \cdots & c_{in} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & c_{mj} & \cdots & c_{mn} \end{bmatrix} \tag{1}$$

Where each row corresponds to the different values of the criteria for a given alternative. Then, c_{ij} represents the value for criterion j in alternative i. A weight vector $w \in \mathbb{R}^n$, with $w_j \geq 0$ and $\sum_j w_j = 1$ needs to be defined with the weights assigned to each criterion and the sum of weights 1.

In the ideal solution, there are some criteria that should be maximized (i.e. productivity or efficiency) and some that should be minimized (i.e. costs or GHG emissions). The *MaxMin* vector contains elements corresponding to each evaluation criterion. The values in this vector (MM_j) are either 1 or -1, where 1 stands for a criterion that should be maximized (a higher value is preferable in the ideal alternative) and -1 stands for the opposite (a lower value for that criterion is preferable in the ideal alternative).

Step 2 - Normalization of the decision matrix (C). Criteria may have different units or scales of measurement, which can make it difficult to compare alternatives. For example, an alternative that has a value of 100 on one criterion could be as good as an alternative that has a value of 5 on another criterion. Normalization of criteria values is a critical step in TOPSIS to address the issue of incommensurability. This normalization process converts criteria to a common scale, ensuring all criteria contribute proportionally without inherent biases due to diverse measurement units or scales. The *standardized decision matrix* (R) is defined in Eq. (2).

$$Stand.Decision\ matrix\ (R) = \begin{bmatrix} r_{11} & r_{12} & r_{1j} & \cdots & r_{1n} \\ r_{21} & r_{22} & r_{2j} & \cdots & r_{2n} \\ r_{i1} & r_{i2} & r_{ij} & \cdots & r_{in} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r_{m1} & r_{m2} & r_{mj} & \cdots & r_{mn} \end{bmatrix} \tag{2}$$

Where r_{ij} is the normalized value of the criterion value c_{ij} calculated with Eq, (3), being MM_j the values 1 or −1 of the *MaxMin* vector defined in Step 1.

$$r_{ij} = \frac{c_{ij}}{\left[\sum_{i=1}^{m} c_{ij}\right]^{1/2}} \cdot MM_j \tag{3}$$

Step 3 – Matrix weighting. This step involves using the weights defined in Step 1 to ensure that each criterion has the desired importance in the evaluation of alternatives. The *Weighted normalized matrix* (F) (Eq. (4)) represents the weighted normalized criterion values of each alternative:

$$Weighted\ normalized\ matrix(F) = \begin{bmatrix} f_{11} & f_{12} & f_{1j} & \cdots & f_{1n} \\ f_{21} & f_{22} & f_{2j} & \cdots & f_{2n} \\ f_{i1} & f_{i2} & f_{ij} & \cdots & f_{in} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ f_{m1} & f_{m2} & f_{mj} & \cdots & f_{mn} \end{bmatrix} \tag{4}$$

Where f_{ij} is the weighted normalized criterion value calculated with Eq. (5) with w_j being the weight assigned to each criterion.

$$f_{ij} = r_{ij} \cdot w_j \tag{5}$$

Step 4 - Determination of positive and negative ideal solutions. The positive-ideal solution (*PIS*) is calculated by selecting the best value for each criterion, while the negative-ideal solution (*NIS*) is obtained by selecting the worst value for each criterion. The positive and negative-ideal values for each criterion can be defined with Eq. (6).

$$\begin{cases} f_j^+ = max\left\{f_{1j}, f_{2j}, f_{3j}, \ldots, f_{mj}\right\} \\ f_j^- = min\left\{f_{1j}, f_{2j}, f_{3j}, \ldots, f_{mj}\right\} \end{cases} \quad (j = 1, 2, 3, \ldots, n) \tag{6}$$

PIS and *NIS* are defined with Eqs. (7) and (8):

$$PIS = \left\{f_1^+, f_2^+, f_3^+, \ldots, f_n^+\right\} \tag{7}$$

$$NIS = \left\{f_1^-, f_2^-, f_3^-, \ldots, f_n^-\right\} \tag{8}$$

Step 5 - Distance calculation to the positive- and negative-ideal solutions. This step assesses the relative proximity of each alternative i to *PIS* (d_i^+) and *NIS* (d_i^-), Eq. (9).

$$\begin{cases} d_i^+ = \sqrt{\sum_{j=1}^{n}\left(f_{ij} - f_j^+\right)^2} \\ d_i^- = \sqrt{\sum_{j=1}^{n}\left(f_{ij} - f_j^-\right)^2} \end{cases} \tag{9}$$

$$Closeness\ Coefficient\ (CC_i) = \frac{d_i^-}{(d_i^+ + d_i^-)} \tag{10}$$

Step 6 – Calculation of the closeness coefficient (CC). CC_i (Eq. (10)) shows how far an alternative i is from the *NIS* solution.

A lower CC_i implies that an alternative i is closer to *NIS* and further away from *PIS*. Conversely, a higher CC_i implies that the alternative i is closer to *PIS* and further away from *NIS*. Therefore, the TOPSIS ranking is established in order of CC_i values from highest to lowest, resulting in a rank of alternatives ordered according to their distance to the negative ideal solution, and equivalently to its proximity to the positive ideal solution.

2.2. Sensitivity analyses

To assess the robustness of the rankings produced by TOPSIS, we applied two complementary sensitivity analysis techniques: a conventional WSA and the novel FRWSA. WSA aims to study the critical weight range for each criterion. This is the extent to which a weight could be increased or decreased (with the remaining weights proportionally adjusted) within which the ranking of the best alternative remains unchanged. For each criterion, we determined the minimum and maximum values that could be assigned without altering the top-ranked option. This method provides insight into the influence of individual criteria on the decision outcome.

In contrast, FRWSA is a multivariate, exhaustive analysis that evaluates the stability of the ranking across the entire feasible space of weight combinations. FRWSA can be viewed as a decision analytics framework that systematically characterizes how preference uncertainty impacts alternative rankings, supporting more informed and robust decision-making. Under the constraint that the sum of weights equals one, FRWSA discretizes the n-dimensional weight simplex into a fine grid and systematically computes the TOPSIS ranking for each valid weight vector. This allows identification of the dominant alternative(s) and the frequency of each alternative being ranked first. Unlike WSA, FRWSA captures interdependencies between criteria and provides a comprehensive view of decision robustness. FRWSA was implemented using a customized MATLAB® script that generates and filters the weight vectors, then applies TOPSIS to each. The implementation details and computational structure of the WSA and FRWSA methods are presented in Sections 3.1 and 3.2.

2.3. Case study

The TOPSIS method was applied to a real case where we previously evaluated the possible retrofitting of a WWTP with nitrogen and organic matter (N/COD) removal to a new configuration with phosphorus, nitrogen and organic matter removal (P/N/COD). Four different plant configuration alternatives were studied [30]: anaerobic/anoxic/oxic with a single anaerobic reactor (A2/O-S) or with two anaerobic reactors (A2/O-D), Bardenpho (BARD) and University of Cape Town (UCT). The WWTP case combines peer-reviewed data with multi-criteria trade-offs of high practical relevance, making it well-suited to weight-sensitivity analysis. In our previous study [31], we evaluated these possible alternatives using data generated with a detailed mechanistic model that was developed and fitted in a full-scale WWTP study [32]. We adopted an engineering approach, normalizing the criteria, building a combined cost function, and performing a basic sensitivity analysis with five different sets of weights. In the present work, the values obtained in the evaluation of ten criteria of that work are used for performing a more exhaustive sensitivity analysis of the four plant alternatives. The set of ten criteria used included crucial aspects of wastewater treatment: capital and operational expenditures (CAPEX and OPEX), removal of organic matter, nitrogen and phosphorus (COD_{REM}, N_{REM} and P_{REM}), average effluent concentration of COD (COD_{AVG}), total N (N_{T-AVG}) and total P (P_{T-AVG}), and robustness test for peaks of P and N (P_{ROB} and N_{ROB}). Table 1 shows the values of the selected criteria for the different alternatives studied and the MaxMin vector.

3. Results and discussion

This section first describes the implementation of the WSA and FRWSA methods. A new interval halving method is implemented to efficiently calculate the WSA ranges. Next, the FRWSA method is designed and implemented, and the results are presented graphically using three different methods. Finally, applying TOPSIS, WSA, and FRWSA to the case study involving four WWTP alternatives provides valuable insights into the comparative performance and robustness of each configuration. This section presents and discusses the outcomes of each analytical approach.

3.1. Implementation of the Weight Sensitivity Analysis (WSA) method

The starting point of WSA is the weighting vector selected for each criterion in the Step 3 of Section 2.1 that leads to a rank of alternatives. The weight of each single criterion is either increased or decreased, while the weights of the remaining criteria are proportionally rescaled to preserve a cumulative weight sum of one. The same procedure is sequentially applied to all the criteria. This analysis finds the range

Table 1
Criteria values obtained for the different alternatives evaluated and MaxMin vector showing the need to maximize (1) or minimize (−1) these criteria.

	1_CAPEX [k€]	2_OPEX [k€]	3_COD_{REM} [%]	4_N_{REM} [%]	5_P_{REM} [%]	6_COD_{AVG} [mg/L]	7_N_{T-AVG} [mg/L]	8_P_{T-AVG} [mg/L]	9_P_{ROB} [kg]	10_N_{ROB} [kg]
(1)-A2/O-S	152.0	2.7	86.2	76.7	77.1	63.4	33.1	7.7	1.8	9.9
(2)-A2/O-D	164.6	2.7	86.1	77.0	82.3	63.8	32.9	4.2	1.6	9.9
(3)-BARD	194.6	12.7	85.9	50.5	70.5	86.0	49.0	6.4	1.4	18.1
(4)-UCT	269.4	3.2	85.8	72.0	86.5	87.3	35.6	4.2	1.3	10.9
MaxMin	−1	−1	1	1	1	−1	−1	−1	−1	−1

of weights for each criterion that does not change the top-ranked alternative obtained with the initial weighting vector.

An interval halving method was developed and run twice for each criterion to find the upper and lower limit of this range. The original interval halving method is a classical iterative method for optimization that helps to reduce the range where the optimum is located. The iterative process is repeated until the desired level of accuracy is reached. Fig. 1 shows a flow chart of the implemented algorithm. The method begins by setting values for the maximum number of iterations and the tolerance. Then, the extreme values of the interval a and b are determined. When aiming at the maximum weight that does not change the top-ranked alternative (Increase Case), a is set to the weight under investigation, and b is set to 1 (since weights range from 0 to 1). Conversely, when finding the minimum weight that does not change the top-ranked alternative (Decrease Case), a is set to 0, and b is set to the weight under investigation. The midpoint weight is calculated and weights of the rest of criteria are proportionally adjusted to maintain the sum of weights to one. TOPSIS is applied to this new weight scenario and the new top-ranked alternative is compared to the initial one.

In the Increase Case, if the midpoint maintains the same ranking of alternatives, the maximum weight will be in the top half. Conversely, if the midpoint ranking changes with respect to the initial, the maximum weight will appear in the lower half. For the Decrease Case, the procedure is the opposite, i.e., if the midpoint maintains the same ranking the upper half does not contain the minimum weight, while if the midpoint changes the top-ranked alternative then the lower half should not be considered. This iterative process continues until either the maximum number of iterations is reached, or the interval value falls below the tolerance. The MATLAB scripts and their description used to implement this algorithm can be found in the supplementary information (*WSA_TOPSIS.m*).

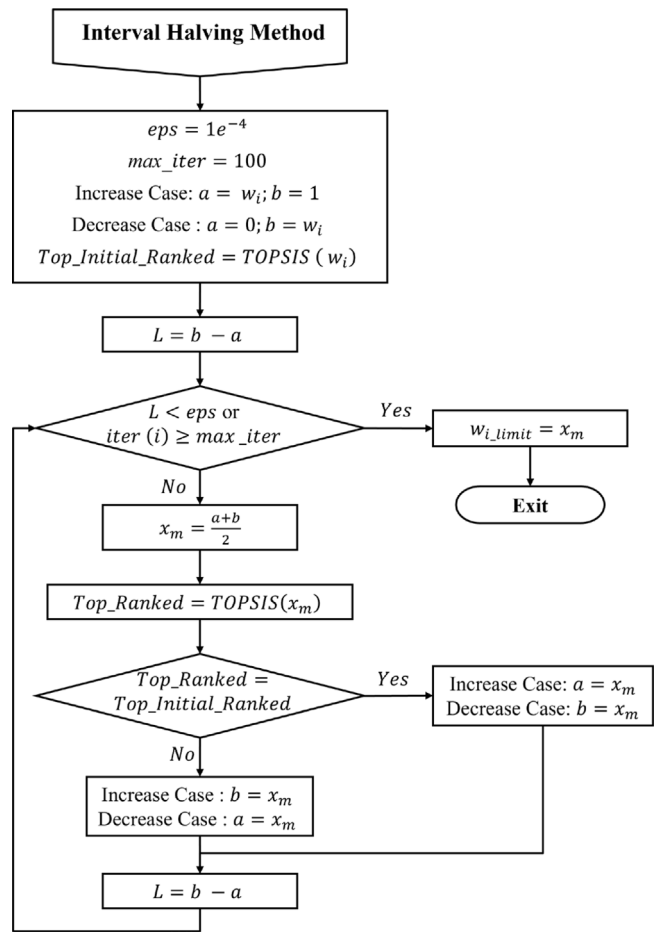

Fig. 1. Flow chart of the interval halving method required in the WSA method.

3.2. Implementation of the full-range weight sensitivity analysis (FRWSA)

FRWSA exhaustively evaluates all feasible weight vectors on a discretized simplex: for a step h, weights take values in $\{0, h, 2h, \ldots, 1\}$ with $\sum_{i=1}^{n.criteria} w_i = 1$, giving $n_W = 1/h+1$ grid levels. Eq. (11) shows an example of the matrix obtained with all grid combinations for a step of 0.05 and with each column referring to the weight of a specific criterion and each row a set of weights that add one, as shown in the last column.

$$Weight\ Matrix\ (WMAT) = \left[\begin{array}{ccccccc|c} 1 & 0 & 0 & 0 & \ldots & 0 & 0 & 1 \\ 0.95 & 0.05 & 0 & 0 & \ldots & 0 & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots & \ddots & 0 & \vdots & \vdots \\ 0 & 0 & 0 & 1 & \ldots & 0 & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \ldots & 0.05 & 0.95 & 1 \\ 0 & 0 & 0 & 0 & \ldots & 0 & 1 & 1 \end{array}\right] \tag{11}$$

The number of rows in the matrix (Eq. (11)) depends on the step size chosen and on the number of criteria. As an example, Table 2 shows all possible combinations for different numbers of criteria studied (from 2 to 10 criteria) with a step of 0.05.

To calculate the total number of valid combinations of weights in the FRWSA method, all discrete weight vectors are considered under the constraint that their components are non-negative and sum to one. Let n be the number of criteria and n_W the number of discrete values in the [0,1] interval. The total number of feasible weight combinations is given by the following combinatorial expression:

$$Number\ of\ combinations = \binom{n_W + n - 2}{n - 1} \tag{12}$$

This expression corresponds to the number of non-negative integer solutions to the equation:

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = n_W - 1, \lambda_i \in \mathbb{N}_0 \tag{13}$$

where each λ_i represents the discretized weight level (step index) for the ith criterion, that is $\lambda_i = w_i/h$, ranging from 0 to $1/h = n_W - 1$. The number of combinations grows rapidly with both the number of criteria and the discretization resolution, as reflected in Table 2.

$$RankingMatrix(RankMAT) = \begin{bmatrix} TOP1(WMAT(1,:)) & TOP1(WMAT(2,:)) & \cdots & TOP1(WMAT(n.Comb,:)) \\ TOP2(WMAT(1,:)) & TOP2(WMAT(2,:)) & \cdots & TOP2(WMAT(n.Comb,:)) \\ TOP3(WMAT(1,:)) & TOP3(WMAT(2,:)) & \cdots & TOP3(WMAT(n.Comb,:)) \\ \vdots & \vdots & \vdots & \vdots \\ TOPn.Alt(WMAT(1,:)) & TOPn.Alt(WMAT(2,:)) & \cdots & TOPn.Alt(WMAT(n.Comb,:)) \end{bmatrix} \quad (15)$$

Box I.

Table 2
Possible combinations for different numbers of criteria studied and using a step of 0.05 ($n_W = 21$).

Number of criteria	*MATLAB function name*	*Number of combinations*
2	Wmatrix_2	21
3	Wmatrix_3	231
4	Wmatrix_4	1,171
5	Wmatrix_5	10,626
6	Wmatrix_6	53,130
7	Wmatrix_7	230,230
8	Wmatrix_8	888,030
9	Wmatrix_9	3,108,105
10	Wmatrix_10	10,015,005

In addition to calculating the total number of feasible weight combinations, it is also useful to determine how many combinations correspond to a specific value of weight assigned to one criterion. This calculation is relevant for constructing frequency plots such as the FRWSA plot described later in this section. When a specific criterion is fixed at a given discretized weight level λ_i, the remaining *n-1* criteria weight levels must distribute the residual sum $n_W - 1 - \lambda_i$ in Eq. (13) among themselves. The number of valid combinations under this condition is given by:

$$Repetitions\left(\lambda_i\right) = \binom{n_W + n - 2 - \lambda_i}{n-2} \quad (14)$$

where i corresponds to the index of the fixed weight level. This expression provides the number of combinations in which one specific criterion takes the value $w_i = \lambda_i \cdot h$, and the remaining weights sum to $1 - w_i$. The result is used to normalize and interpret the frequency with which an alternative becomes top-ranked for different individual weight values.

Detailed descriptions of the MATLAB scripts used to implement this method can be found in the supplementary information (*FRWSA_TOPSIS.m*). The implementation was validated via: (i) synthetic matrices with known PIS/NIS to unit-test closeness coefficients; (ii) a small case enumerated analytically and by code; and (iii) integrity checks ensuring that, for each fixed weight level in FRWSA strips, the sum of rank-1 frequencies across alternatives equals one.

Once all the combinations are calculated, TOPSIS is applied to each case. The result of applying the TOPSIS method to each scenario is the Ranking Matrix (*RankMAT*) (Eq. (15)), where each column of *RankMAT* corresponds to the ranking results for a combination of weights considered, i.e. each row of *WMAT* (Eq. (11)). The first row contains the index of the highest ranked alternative, the second row contains the index of the TOP 2 and subsequent rows progressively list the remaining alternatives in descending order until the lowest ranked alternative in row n. Therefore, the matrix dimensions shall correspond to the number of alternatives (n_Alt) for the rows and the total number of combinations of weights for the columns (n_Comb). For example, in the case of 4 alternatives and 6 different criteria, the matrix will have 4 rows and 53,130 columns (Table 2). See Eq. (15) in Box I. With all rankings for all possible scenarios, the results are presented through three different graph types to better communicate these insights.

- Pie plot: Represents the fraction of scenarios that each alternative is the top-ranked. For this purpose, the frequency of each alternative appearing in the first row of *RankMAT* is calculated.
- Frequency distribution plot of the ranking position: This heat map shows the frequency of each alternative in each ranking position.
- FRWSA plot: It is a histogram graph plot for each alternative showing, the fraction of scenarios in which the alternative (e.g., *k)* is the best option for each criterion *j* and specific weighting value *i*.

By identifying the position of each alternative, it is possible to determine the combinations of weights in which that alternative is the best option, thus resulting in the *WRank1Alt*(*k*) (Eq. (16)).

$$WRank1Alt(k) = WMAT(columnRank1,:) \quad (16)$$

The matrix *FRMAT* (Eq. (17)) contains the fraction of all possible cases in which alternative *k* is top-ranked for each criterion *j* and specific weighting value *i, i.e.* the dominance frequency frc_{ijk} (Eq. (17)).

$$FRMAT(i,j,k) = \begin{bmatrix} frc_{11k} & frc_{12k} & frc_{13k} & \cdots & frc_{1jk} \\ frc_{21k} & frc_{22k} & frc_{23k} & \ldots & frc_{2jk} \\ frc_{31k} & frc_{32k} & frc_{33k} & \ldots & frc_{3jk} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ frc_{i1k} & frc_{i2k} & frc_{i3k} & \ldots & frc_{ijk} \end{bmatrix} \quad (17)$$

The dimension of *FRMAT* varies according to the values that a weight can take, and the number of criteria studied, i.e., if the values of the weights range from 0-1 divided by steps of size 0.05 (21 different values) and 6 criteria, the matrix will be 21 rows and 6 columns for each alternative (*k*). A value $frc_{ijk} = 0$ means that alternative *k* is never the best for a weight w_{ij} of criterion *j*. Conversely, $frc_{ijk} = 1$ means that this alternative *k* is always the best if we use the weight w_{ij} of criterion *j*. Moreover, the sum of the frc_{ijk} for all alternatives in a specific weight w_{ij} is always one ($\sum_{k=1}^{n.Alt} frc_{ijk} = 1$).

3.3. TOPSIS results with fixed weights

TOPSIS was first applied using a fixed set of equal weights of 0.1 as a benchmark due to its simplicity and neutrality, although weight values are typically derived from expert judgment and literature. The results of the closeness coefficient, distance to the positive and negative ideal solutions and the TOPSIS ranking are presented in Table 3. The closeness coefficients are shown in Fig. 2 while the distances to PIS and NIS are presented in the SI (Figures S2 and S3).

The optimal alternative found was (2)-A2/O-D, followed by (1)-A2/O-S, (4)-UCT and, finally, (3)-BARD. It is important to highlight that (2)-A2/O-D is found to be the best alternative with the combination of weights (equal weights $w = 0.1$), while this ranking could change with a different combination of weights. The fixed-weight approach provided an initial benchmark but did not account for the potential variability in decision-maker preferences. As an additional example, a second scenario using an expert-elicited weight vector is included in SI

Table 3
TOPSIS results and ranking of alternatives obtained with TOPSIS using a weight of 0.1 for the 10 criteria.

Alternative	Description	Closeness coefficient	Distance to PIS d_i^+	Distance to NIS d_i^-	Ranking
(1)-A2/O-S	Anaerobic/anoxic/oxic (A2/O) configuration with a single anaerobic reactor	0.7245	0.0347	0.0912	2
(2)-A2/O-D	Anaerobic/anoxic/oxic (A2/O) configuration with two anaerobic reactors	0.9003	0.0106	0.0956	1
(3)-BARD	Bardenpho configuration	0.2206	0.0896	0.0254	4
(4)-UCT	University of Cape Town configuration	0.7174	0.0341	0.0864	3

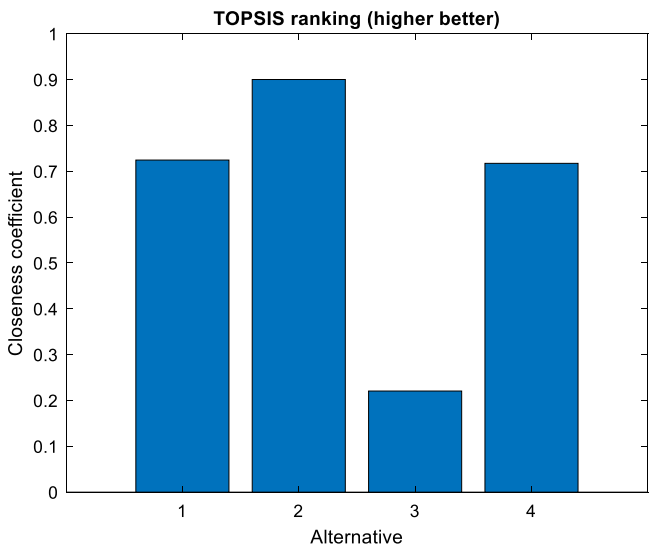

Fig. 2. TOPSIS Closeness Coefficients (CC) for the four alternatives studied. Alternative (2)-A2/O-D clearly gives the highest value and is therefore the best option considering the combination of weights used (equal weight $w_i = 0.1$). The range of possible values for CC is 0–1.

(Table S1 in Section S3), in which the total weight of the economic criteria is increased by up to 50%. The TOPSIS results in Table S2 and Figures S4-S6 show that (2)-A2/O-D remains the best alternative. However, (1)-A2/O-S achieves similar values in this scenario, making it a more attractive option due to its lower CAPEX.

3.4. Sensitivity analysis with WSA

WSA studies the sensitivity of the ranking of alternatives (Table 3) to variations in the weight assigned to each criterion following the procedure detailed in Section 3.1. Thus, the range of the values of a single weight that did not affect the top-ranked alternative was evaluated. Fig. 3 (and Table S3) shows the range that each criterion can take without moving (2)-A2/O-D as the best ranked alternative. For example, several weights (2_OPEX, 4_N_{REM}, 7_N_{T_AVG}, 8_P_{T_AVG} and 10_N_{ROB}), can range from 0 to 1 without affecting the best alternative status of (2)-A2/O-D. An upper limit of one is expected for these criteria, as alternative (2)-A2/O-D is the best one according to each of these criteria. An increase in certain weights (1_CAPEX, 3_COD_{REM}, 5_P_{REM}, 6_COD_{AVG} and 9_P_{ROB}) makes the (2)-A2/O-D alternative not the best choice. For instance, the maximum weight for the robustness test for P (9_P_{ROB}) was 0.27. That means that values higher than 0.27 would result in (2)-A2/O-D not being the best ranked alternative because other configurations perform better on this criterion.

Table 4 shows examples of TOPSIS results for some weight scenarios calculated with slightly out-of-range values according to Fig. 3. For example, when 1_CAPEX weight is set to 0.53, (1)-A2/O-S becomes

Table 4
Results of the best alternative change with the increase in weight values.

Criterion	*Criterion value*	*Ranking*	*Best alternative*
1_CAPEX	0.53	[1, 2, 3, 4]	(1)-A2/O-S
3_COD_{REM}	0.99	[1, 2, 4, 3]	(1)-A2/O-S
5_P_{REM}	0.56	[4, 2, 1, 3]	(4)-UCT
6_COD_{AVG}	0.94	[1, 2, 3, 4]	(1)-A2/O-S
9_P_{ROB}	0.28	[4, 2, 1, 3]	(4)-UCT

the top alternative, since (1)-A2/O-S offers a lower 1_CAPEX (152.0 k) than (2)-A2/O-D (164.6 k€). Regarding 9-P_{ROB}, when the weight is set to 0.28, the (4)-UCT configuration is found to be the optimal alternative, since (4)-UCT gives better results than (1)-A2/O-S in this criterion (1.3 kg vs 1.8 kg).

Overall, the WSA revealed that, in this case, (2)-A2/O-D is the best alternative for a wide range of selected weights, making it a robust choice as the best performer. (2)-A2/O-D retains its top position unless one of these criteria becomes extremely important to the user and is therefore assigned a very high weight outside the maximum range.

3.5. Global sensitivity with FRWSA

The FRWSA offered a more comprehensive analysis by evaluating all feasible weight combinations for the 10 criteria. The weights were calculated for 21 different levels (from 0 to 1 at 0.05 steps). Then, a total of 10,015,005 combinations (Table 2) were evaluated. This analysis assesses how often a configuration is the best with respect to the entire n-dimensional space of possible weight combinations. Fig. 4 shows the percentage distribution of the top-ranked alternative across the full grid. The (2)-A2/O-D alternative exhibits a remarkable dominance, being the top-ranked in 83% of the cases. This alternative obtains the best results compared to the other alternatives in most of the criteria. (4)-UCT is the second best (12%), (1)-A2/O-S the third best (5%) and finally, (3)-BARD is the least often top-ranked, occupying the first position with the lowest frequency, as it gives the worst results in most of the criteria.

Fig. 5 shows a heat map summarizing the frequency distribution of alternative positions under various weight combinations. The y-axis represents the alternatives considered, while the x-axis represents their ranking positions. The alternative (2)-A2/O-D exhibits a prominent presence in the top rank and frequently secures the second and sometimes the third position, but never occupies the last position, which demonstrates its superior performance. The next most successful alternative is (4)-UCT, which proves to be the second-best alternative in first positions, but also appears frequently as second or third best option. In contrast, the alternative (3)-BARD is the bottom-ranked alternative and systematically appears as the worst option, although in a small number of specific cases it may be the first choice.

Fig. 6 shows 10 subplots for the best alternative, (2)-A2/O-D, where for each criterion j and specific weight value i (w_{ij}), the fraction of cases where (2)-A2/O-D is the best option is represented (frc_{ij2} dominance

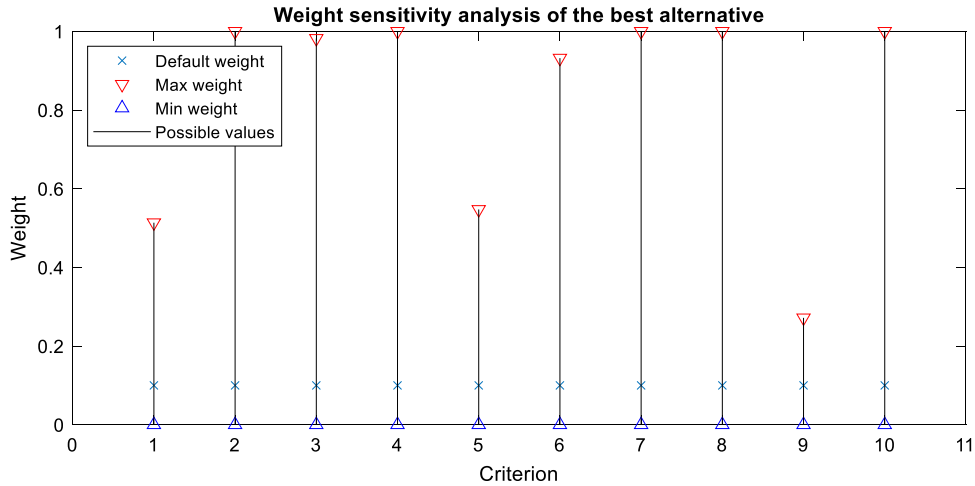

Fig. 3. Weight sensitivity analysis (WSA) results for the selection of WWTP alternatives. Default weight represents the original weights used that resulted in the selection of (2)-A2/O-D as the best alternative. The line indicates all the weight values obtained following the procedure detailed in Section 3.1 that maintain (2)-A2/O-D as the best choice, with Max and Min weight as the limiting values for these ranges.

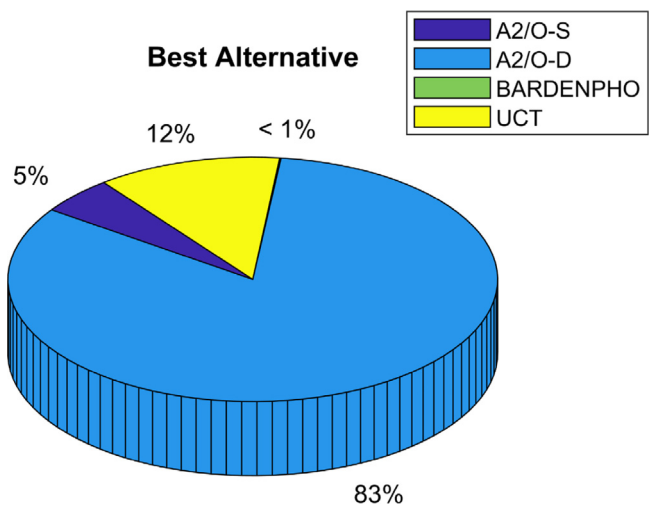

Fig. 4. Percentage distribution of the first position of each alternative obtained with the FRWSA-TOPSIS method.

frequency). The same complete graphs for the other alternatives can be found in the SI (figures S7-10).

Fig. 6 offers valuable insights into the prevalence of weights favouring the top ranking of alternative (2)-A2/O-D ($k = 2$). For example, the first subplot shows the effect of the weight selected for CAPEX (criterion 1, w_{i1}) on the dominance frequency related to alternative 2 (frc_{i12}). The dominance frequency frc_{i12} exhibits a predominantly high value (near one) for many weight ranges. However, as w_{i1} is increased, (2)-A2/O-D progressively loses relevance as the top-ranked option. Other criteria related to N removal such as 4_N_{REM} (Criterion 4, w_{i4}) and 7_N_{T-AVG}(Criterion 7, w_{i7}) are even more favourable, since they have frc_{i42} and frc_{i72} values near one for the whole weight range 0–1, and they only give some chances to other alternatives for low weight values.

Fig. 7 compares the four alternatives with the results of the FRWSA-TOPSIS of the 3_COD_{REM} criterion (w_{i3}). The *frc* dominance frequency results for (2)-A2/O-D (frc_{i32}) show that it is the best alternative across the weight value range. This prevalence is more significant for lower weight values, and it decreases when exceptional emphasis is placed on a single criterion (value close to 1). For instance, only if 1_CAPEX becomes very relevant (i.e. a high weight is placed), (1)-A2/O-S replaces (2)-A2/O-D as the top-ranked alternative. (1)-A2/O-S also emerges as the optimal alternative in this single criterion due to its superior performance in terms of COD reduction value (86.2% COD removal). On the other hand, it is observed that the (3)-BARD alternative offers the worst results in the sensitivity analysis (frc_{i33}), although the (4)-UCT (frc_{i34}) is the worst alternative for this criterion (85.8% COD removal). Overall, the (3)-BARD alternative offers worse results combining the other criteria, being the alternative that appears less frequently as the best one. Finally, Fig. 7 also allows to confirm that the sum of all dominance frequency *frc* for a given weight value gives always one, i.e., for each weight value w_{i3} it can be checked that $frc_{i31} + frc_{i32} + frc_{i33} + frc_{i34} = 1$.

3.6. Computational performance and comparison with Monte Carlo methods

The computational analysis was carried out using a personal computer equipped with an Intel® Core™ i7-1255U processor (1.7 GHz), 32 GB of RAM (3200 MHz), integrated Iris® Xe Graphics, a 1 TB SSD, and running Windows 11 Home 64-bit. The MATLAB® code implementing the TOPSIS, WSA, and FRWSA methods was executed ten times for three different weight discretization levels to evaluate performance consistency. The results are summarized in Table 5.

For the reference case using a weight step of 0.05 – corresponding to approximately 10 million evaluations of the TOPSIS method – the average computation time was only 104 s. When the step was increased to 0.1, the number of combinations decreased significantly to 92,378, resulting in an average execution time of just 3 s. Conversely, reducing the step to 0.04 increased the total combinations to over 52 million, with the execution time rising accordingly to 319 s.

Despite the large number of combinations assessed, the FRWSA method demonstrated computational feasibility, efficiency and practicality. Even at high resolutions, the simulation times remained within a practical range for modern computing systems. These results confirm that the exhaustive nature of FRWSA does not entail prohibitive computational costs, making the method viable for real-world decision-support applications.

MC methods are widely used for exploring complex stochastic spaces and, when properly used, allow to reduce the number of total evaluations required to obtain a good approximation of the exact result. In a common MC approach, weights are constructed randomly generating valid weight vectors and normalizing them to guarantee $\sum_{i=1}^{n.criteria} w_i = 1$. However, this sampling procedure can introduce undesirable biases since uniformly normalized random weights tend to

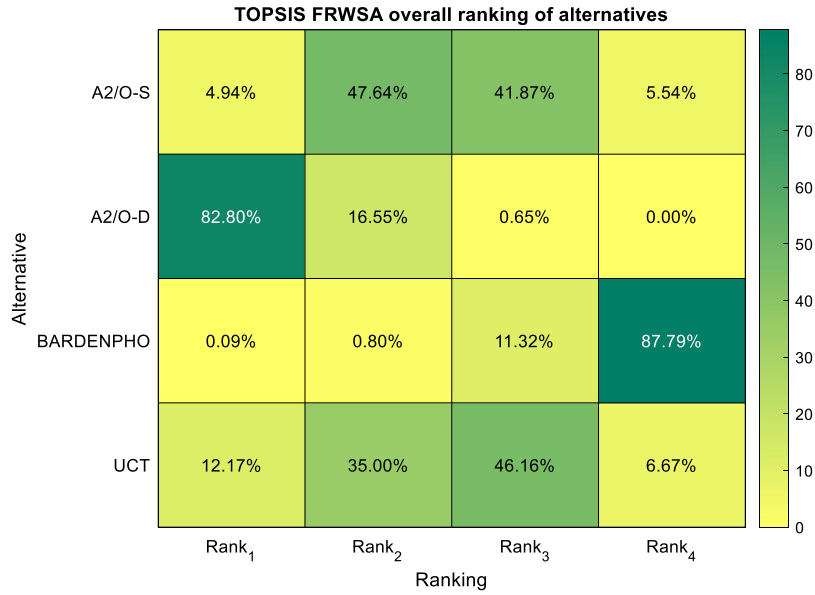

Fig. 5. FRWSA-TOPSIS frequency distribution of the ranking position for each alternative.

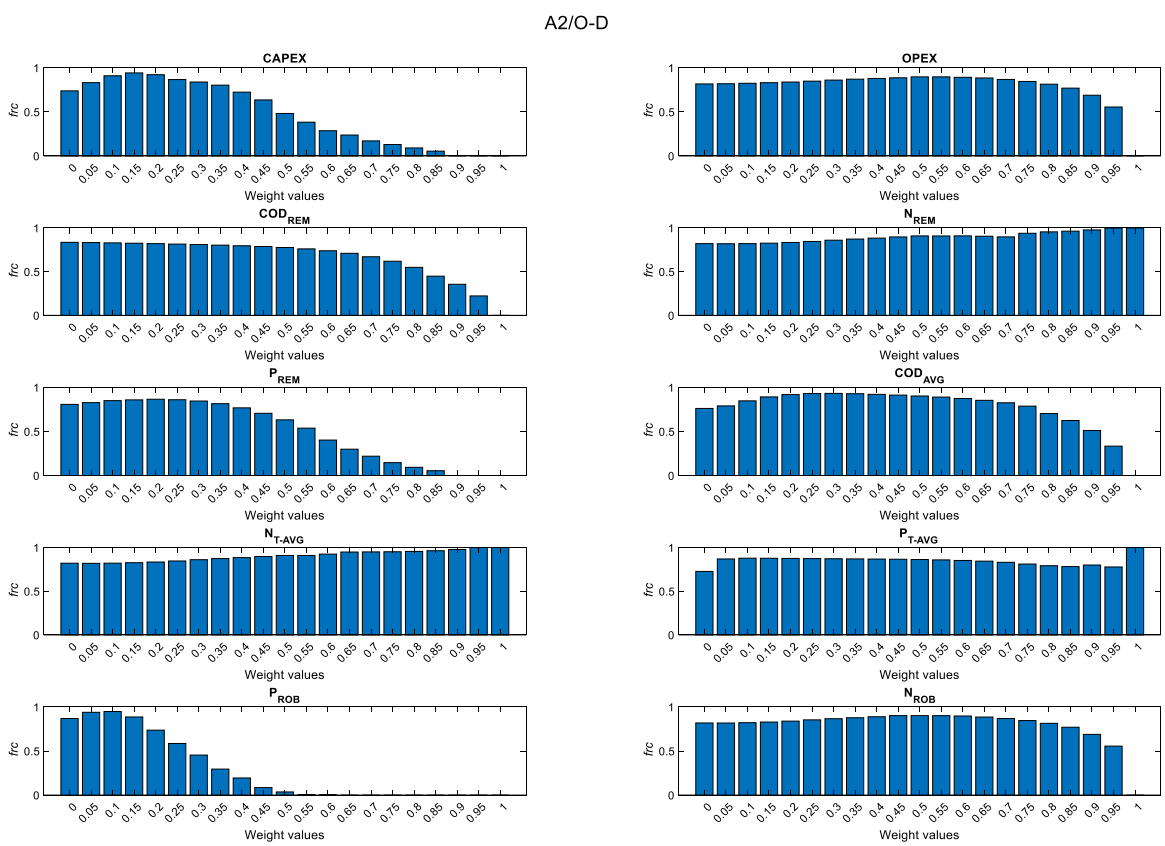

Fig. 6. Results of the FRWSA-TOPSIS dominance frequency for alternative (2)-A2/O-D. Each subplot represents, for a given criterion *j*, the fraction of combinations of weights of the other criteria where (2)-A2/O-D is the best ranked alternative when the specific weight value indicated on the *x*-axis for that criterion *j* is used.

Table 5
Effect of weight discretization on the number of combinations and average execution time for ten runs of TOPSIS, WSA, and FRWSA.

Weight step	Number of combinations	Average execution time (s)
0.1	92 378	3.1 ± 0.1
0.05 (case study)	10,015,005	104.4 ± 0.6
0.04	52,451,256	319.4 ± 2.9

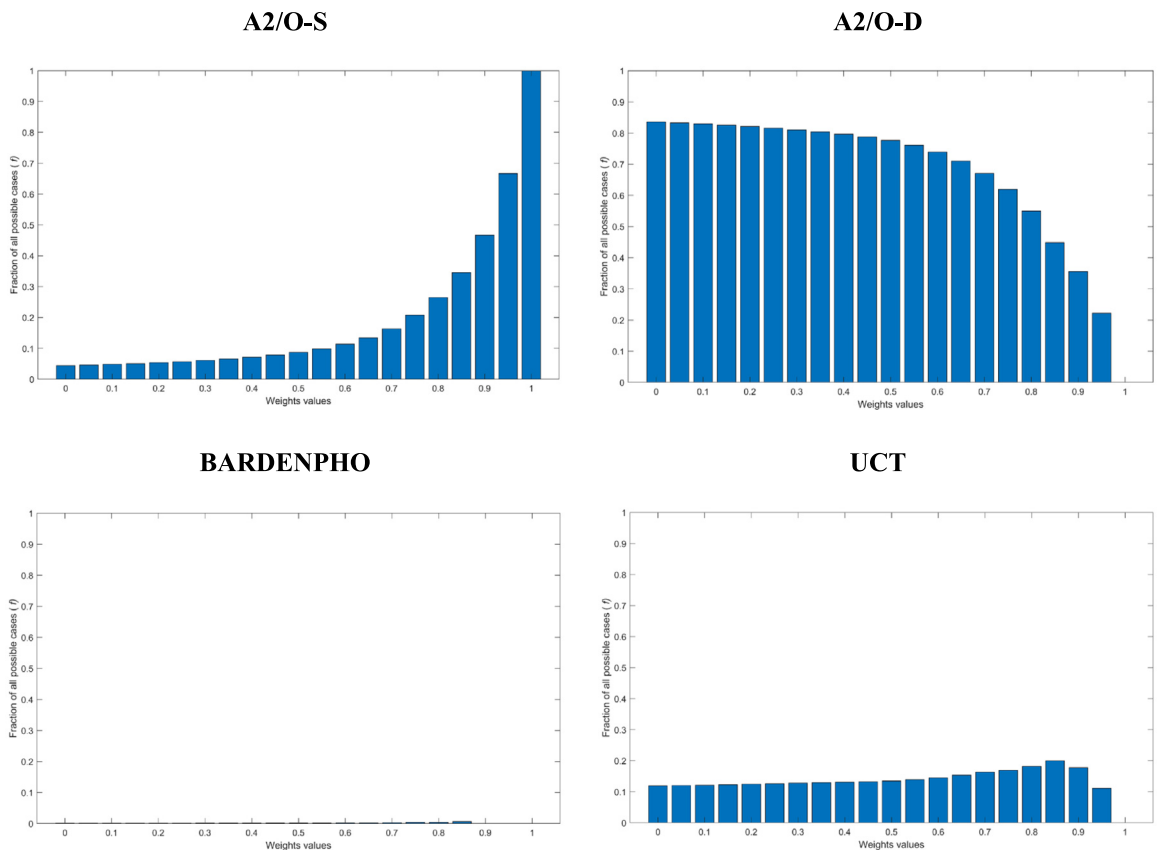

Fig. 7. FRWSA-TOPSIS analysis of the weight 3_COD$_{REM}$ for alternatives (1)-A2/O-S, (2)-A2/O-D, (3)-BARD and (4)-UCT.

underrepresent extreme weight combinations. The number of points in the weighting space near the corners (weight close to 1) is very small, so even uniform sampling will rarely produce such extreme points unless the number of samples is quite large. As a result, combinations of criteria in which a single criterion predominates (scenarios that can alter the TOPSIS ranking) may be overlooked or insufficiently sampled. Fig. 8a shows the empirical distribution of the weights under this normalization (generated with $N = 10^5$ simulations): where points grouped at the vertices are systematically underrepresented.

As an alternative, Dirichlet sampling [33,34] generates valid weight vectors and allows dispersion to be controlled. However, MC–Dirichlet (MC–D) introduces an important practical choice, alpha (α) value: a small value ($\alpha < 1$) intentionally emphasizes the extremes (improving the distribution towards the corners), while an $\alpha = 1$ or higher produces a distribution further away from these extremes of the weight space. In Fig. 8(b-d), different α values produce visibly different weight distributions; therefore, this value creates a degree of freedom that modifies the weight dispersion and can affect the ranking results. α should be selected to best suit the desired dispersion, otherwise MC–D results may overrepresent (i.e. $\alpha = 0.1$) or underrepresent (i.e. $\alpha = 1.0$) extreme scenarios. This distortion may affect the accuracy of the results and reduce their practical relevance.

The agreement between MC dominance profiles and the FRWSA baseline was evaluated using the Jensen–Shannon divergence (JSD, bits, base-2), its square-root distance $d_{JS} = \sqrt{JSD}$, the Euclidean norm ($L_2 = \| P - Q \|_2$), and total variation ($TV = \frac{1}{2} \| P - Q \|_1$) (Table 6). *N. Sim* denotes the number of random weight draws in MC/MC–D and the number of grid evaluations at step *h* for FRWSA. Time is reported in elapsed seconds and MC values report mean ± SD over n=5 runs at each *N*, similarity metrics were computed on the mean dominance vector per row; run-to-run SDs were small relative to mean values; FRWSA rows are deterministic.

Sampling weights without a Dirichlet parameterization (MC case) leads to a dominance profile that remains at a roughly constant distance from the FRWSA baseline as number of simulations increases: *JSD* ≈0.043–0.044 bits (d_{JS}≈0.208), L_2≈0.171, *TV*≈0.137 for 10^4–10^7 simulations. The stability of these values across sample sizes indicates a systematic mismatch due to the underlying weight-sampling scheme rather than the MC variance.

Using the symmetric MC–D with $\alpha = 1.0$ is a principled default when no additional information about plausible weight patterns is available. In this setting, distances to the FRWSA baseline are small but persistent: *JSD* ≈0.0075–0.0078 bits (d_{JS}≈0.087–0.088), L_2≈0.073, *TV*≈0.060, essentially unchanged from 10^4 to 10^7 simulations. Differences are therefore attributable to the choice of continuous uniform measure over the simplex, not to insufficient sampling. MC–D with α=0.5 places more weights near edges/corners of the simplex and, in this case, yields very close agreement with the FRWSA baseline: *JSD* ≈0.0010–0.0011 bits (d_{JS}≈0.032), L_2≈0.036–0.038, *TV*≈0.029–0.030, with rapid stabilization of dominance percentages as number of simulations increases.

Regarding the computational cost and grid sensitivity, elapsed time scales with the number of evaluations. FRWSA requires 3.1 s at $h = 0.10$ ($9.24 \cdot 10^4$ evaluations), 105 s at $h = 0.05$ ($1.00 \cdot 10^7$), and 317 s at $h = 0.04$ ($5.25 \cdot 10^7$). Refining the grid from $h = 0.05$ to 0.04 produces only minor shifts (*JSD* =0.0003, $d_{JS} = 0.018$, $L_2 = 0.0156$, $TV = 0.0126$). MC/MC–D times grow more gently with number of simulations (e.g., 1.4–1.9 s at 10^4–10^5; 56–62 s at 10^7). For a target of near-identical dominance frequencies relative to the $h = 0.05$ grid, MC–D with $\alpha = 0.5$ attains *JSD* ≈0.001 in ~7–62 s (10^6-10^7 draws),

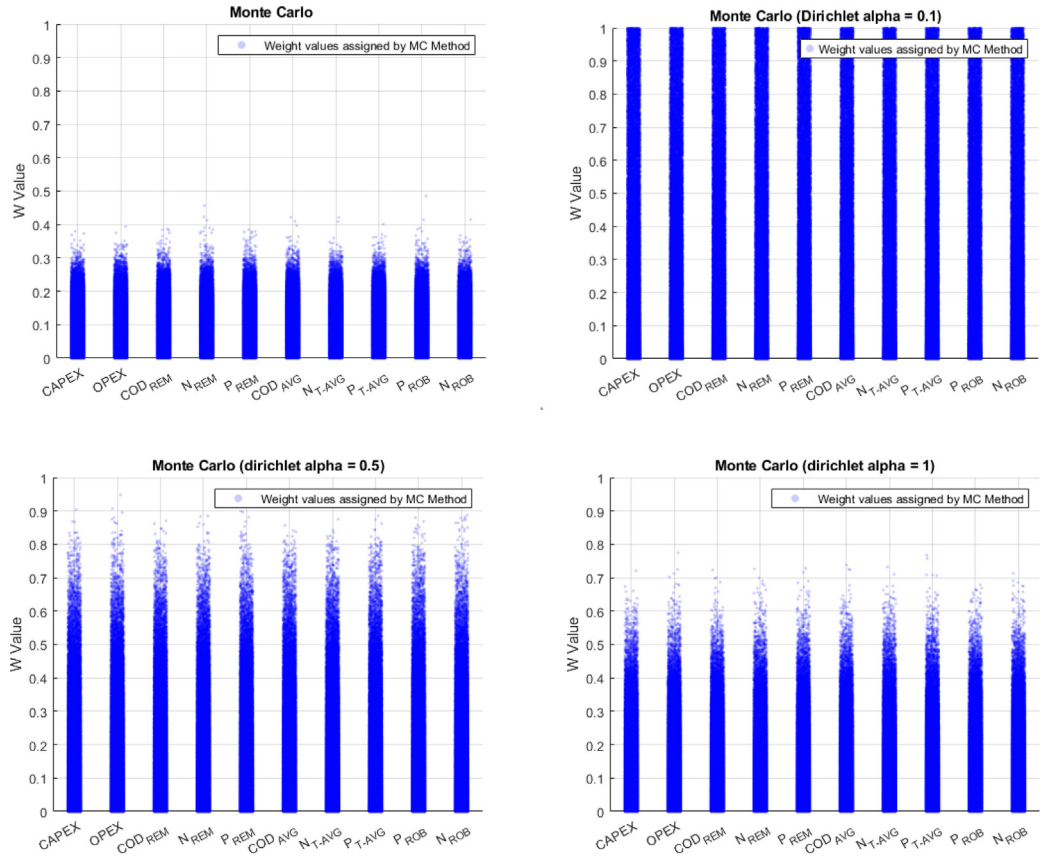

Fig. 8. Distribution of weight vectors using 10^5 simulations. Monte Carlo sampling (a), and MC–Dirichlet with $\alpha = 0.1$ (b), $\alpha = 0.5$ (c) and $\alpha = 1.0$ (d).

Table 6
Agreement between dominance-frequency vectors produced by plain Monte Carlo (MC), MC–Dirichlet (MC–D; parameter α), and FRWSA for different numbers of evaluations. For MC/MC–D, *N. Sim* is the number of random weight draws; for FRWSA it is the total grid evaluations at step h. Reported are elapsed time (s) and dominance percentages for [A2/O-S, A2/O-D, BARD, UCT]. Similarity to the FRWSA baseline at $h = 0.05$ is quantified by JSD (bits, base-2), JSD distance d_{JS}, Euclidean norm ($L_2 = \| P - Q \|_2$), and total variation ($TV = \frac{1}{2} \| P - Q \|_1$); lower values indicate closer agreement. MC and MC–D rows show mean ± SD (n = 5 independent runs) at each *N. Sim*; FRWSA rows are deterministic. Similarity metrics are computed using the mean dominance vector for each row.

Method	*N. Sim*	*Time (s)*	*A2/O-S (%)*	*A2/O-D (%)*	*BARD (%)*	*UCT (%)*	*JSD (bits)*	d_{JS}	L_2 *norm*	*Total variation*
MC	1.0E4	1.45 ± 0.14	0.18 ± 0.03	96.48 ± 0.26	0.00 ± 0.00	3.34 ± 0.24	0.0434	0.2084	0.1696	0.1368
MC	1.0E5	1.89 ± 0.30	0.22 ± 0.01	96.54 ± 0.05	0.00 ± 0.00	3.24 ± 0.06	0.0434	0.2083	0.1706	0.1374
MC	1.0E6	6.40 ± 0.25	0.21 ± 0.01	96.57 ± 0.02	0.00 ± 0.00	3.21 ± 0.02	0.0437	0.209	0.1709	0.1377
MC	1.0E7	55.79 ± 1.18	0.22 ± 0.00	96.57 ± 0.01	0.00 ± 0.00	3.22 ± 0.01	0.0436	0.2087	0.1709	0.1377
MC–D (α=0.5)	1.0E4	1.38 ± 0.12	5.53 ± 0.25	79.83 ± 0.35	0.18 ± 0.05	14.47 ± 0.30	0.0011	0.0338	0.038	0.0297
MC–D (α=0.5)	1.0E5	2.14 ± 0.11	5.72 ± 0.05	79.89 ± 0.11	0.15 ± 0.02	14.23 ± 0.07	0.001	0.0322	0.0365	0.0291
MC–D (α=0.5)	1.0E6	7.45 ± 0.61	5.68 ± 0.01	79.93 ± 0.03	0.16 ± 0.00	14.22 ± 0.03	0.001	0.0319	0.036	0.0287
MC–D (α=0.5)	1.0E7	62.26 ± 0.63	5.68 ± 0.01	79.92 ± 0.01	0.17 ± 0.00	14.24 ± 0.01	0.001	0.0323	0.0362	0.0288
MC–D (α=1.0)	1.0E4	1.50 ± 0.04	1.83 ± 0.10	88.82 ± 0.32	0.04 ± 0.03	9.31 ± 0.28	0.0075	0.0869	0.0735	0.0602
MC–D (α=1.0)	1.0E5	2.32 ± 0.16	1.81 ± 0.04	88.75 ± 0.06	0.03 ± 0.01	9.42 ± 0.07	0.0076	0.0871	0.0726	0.0594
MC–D (α=1.0)	1.0E6	8.62 ± 0.26	1.77 ± 0.01	88.79 ± 0.02	0.03 ± 0.00	9.41 ± 0.03	0.0078	0.0881	0.0732	0.0599
MC–D (α=1.0)	1.0E7	61.39 ± 1.50	1.77 ± 0.00	88.78 ± 0.01	0.03 ± 0.00	9.42 ± 0.01	0.0078	0.0881	0.073	0.0598
FRWSA (h = 0.1)	9.24E4	3.1	9.52	76.07	0.18	14.23	0.007	0.0839	0.084	0.0673
FRWSA (h = 0.05) (baseline)	1.00E7	104.95	4.94	82.80	0.09	12.17	0	0	0	0
FRWSA (h = 0.04)	5.25E7	317.48	4.14	84.06	0.07	11.73	0.0003	0.0181	0.0156	0.0126

whereas FRWSA at $h = 0.05$ takes ~105 s but is exact on the chosen grid and variance-free. Coarser FRWSA (h = 0.10) completes in ~3 s at the cost of larger deviations (*JSD* ≈0.007). For larger n or finer h, FRWSA supports coarse-to-fine enumeration (start with larger h, refine near observed decision boundaries) and stratified slices that fix one weight and scan the residual simplex (used in FRWSA strips). This can preserve interpretability while managing cost.

MC–D requires the selection of a proper α, different α values change the dispersion of the weights and therefore the frequencies of the results. When FRWSA is not available, reporting results for $\alpha = 1.0$ together with a brief α-sensitivity check (e.g., $\alpha \in \{0.5,1,2\}$) can help to qualify conclusions without extensive tuning. Where feasible, FRWSA avoids the need to select α altogether and provides a deterministic, variance-free benchmark at the chosen grid resolution.

The FRWSA method introduced in this study, deterministically and systematically generates all feasible weight vectors on a discretized simplex, thus covering the entire feasible weight space without bias. Each possible weighting scenario is examined exactly once, and its results are recorded. This exhaustive approach ensures that even the most distant extremes of the weighting space are included in the analysis and counted in the correct proportion. Therefore, FRWSA provides fully reproducible and unbiased sensitivity results. Compared with alternatives: (i) WSA is fast and intuitive but one-at-a-time and misses interactions; (ii) MC/MC–D approximate continuous measures with tuneable cost but introduce sampling variance and dependence on α; (iii) Unlike visual/simulation approaches that sample or illustrate subsets of the simplex, FRWSA is deterministic and exhaustive on the chosen grid, yielding variance-free dominance frequencies at the cost of combinatorial growth with n and finer steps. FRWSA scan also supports multi-objective settings by mapping scalarized weights to Pareto regions; coarse-to-fine grids can highlight robust regions of the Pareto front for subsequent detailed optimization.

3.7. Comparison with other weight selecting methods

The results align with prior evidence that TOPSIS rankings can shift markedly with the chosen weight vector (e.g., [19,35]). In this case, dominance-frequency maps show that A2/O-D is top-ranked across most admissible weights on the discretized simplex (≈83% at h = 0.05; ≈84% at h = 0.04), providing a quantitative notion of robustness rather than a single "best" weighting (Table 6). Compared with one-at-a-time WSA [20], FRWSA explores the entire discretized simplex and captures multi-criterion interactions; per-criterion strips make decision-boundary shifts explicit.

Closed-form weight stability intervals (WSI) for the weighted-sum model [36] provide exact ranges of weights that preserve full or partial rankings, and identify sensitive criteria without enumeration; this analytic view complements FRWSA's exhaustive mapping of the weight simplex for distance-based cores, which yields dominance frequencies and explicit decision-boundary structure.

Robust TOPSIS variants address non-monotonic preferences, hierarchical structures, and normalization choice, summarizing outcomes via stochastic multicriteria acceptability analysis [37]; such designs strengthen modelling fidelity but do not quantify coverage of the entire weight simplex. FRWSA is complementary by characterizing weight-space robustness directly through simplex enumeration.

Visual approaches improve explainability by showing how weights and aggregation rules affect TOPSIS rankings [22], but they do not enumerate the full weight simplex or quantify coverage (e.g., the share of admissible weights for which an alternative dominates). Recent studies [38] assess ranking stability via Monte-Carlo perturbations and metaheuristic weight tuning (e.g., particle swarm optimization), reporting method-dependent variability and design sensitivity. FRWSA's grid provides a global robustness map and can serve as a variance-free baseline to calibrate such simulations (e.g., via $JSD/L_2/TV$) and to separate sampling effects from intrinsic weight-space structure. On the simulation side, sensitivity analyses remain prevalent [29], but conclusions depend on the sampling design and parameterization. Here, the agreement study shows that MC–D with α =1 .0 (uniform on the simplex) yields small but persistent deviations from FRWSA grid (JSD ≈0.007–0.008 bits), α = 0.5 yields near-identity (JSD ≈0.001 bits), and plain MC remains further away (JSD ≈0.043 bits) across 10^4-10^7 draws, underscoring that sampling-based conclusions are α- and design-conditional, whereas FRWSA provides a deterministic benchmark.

Objective/data-driven schemes such as ROCOSD embed robustness into weight estimation (maximizing the minimal weight change needed for rank reversal while balancing dispersion/correlation) [39]. Other related works derive weights from data, e.g., maximizing-deviation weighting in an NWPHF-TODIM setting [23], entropy/Schweizer-Sklar-based aggregation under unknown weights in a circular q-rung framework [24], a q-ROF 2-tuple VIKOR+QUALIFLEX scheme with partial-information weights [25], and a cubic q-ROF TODIM with Minkowski-type distances and entropy-based weighting [26]. FRWSA does not choose a single vector; it quantifies how widely any proposed weights are supported in the simplex. Dominance-frequency and boundary maps indicate whether conclusions persist over broad preference regions or rely on narrow regimes, and can guide interactive interfaces towards stable areas [21].

FRWSA is a method-agnostic approach to reporting ranking transparency and robustness: it can wrap alternative cores (e.g., VIKOR, PROMETHEE, TODIM) while maintaining the same weight-space coverage. For example, FRWSA could be applied to other MCDM methods that require a similar procedure for defining weights, such as the VIKOR method. Figures S11-S17 show the results obtained by applying FRWSA-VIKOR. The results are comparable to those obtained using FRWSA-TOPSIS, indicating that robustness conclusions stem from the weight exploration rather than a specific ranking core. The evolving software packages such as decideXpert [40], PySensMCDA [41] and pyfdm [42], underscore demand for reproducible sensitivity workflows; FRWSA aligns with this trend by offering a transparent, method-agnostic grid-based procedure that can be integrated into such tools.

Limitations of FRWSA-TOPSIS include the dependence on the chosen grid resolution (finer steps increase cost), and combinatorial growth for large n. Dominance frequencies are exact on the grid but approximate the continuous simplex. Complementary MC–D analyses quantify residual differences, and natural extensions include broader cross-domain benchmarks and adaptive/coarse-to-fine grids.

4. Conclusions

This work introduces FRWSA to obtain a transparent, deterministic view of ranking robustness under weight uncertainty. In the WWTP selection case, A2/O-D is top-ranked across most admissible weights on the discretized simplex (≈82.8% at h = 0.05 and ≈84.1% at h = 0.04) with only small changes under grid refinement, providing a quantitative notion of robustness beyond any single weight vector. These findings suggest that A2/O-D configuration offers a reliable balance between economic, environmental, and technical performance. A head-to-head analysis clarifies the relation to sampling. Plain MC remains at a roughly constant distance from the FRWSA baseline (JSD ≈0.043 bits) across 10^4-10^7 draws; MC–D α = 1.0 (uniform on the simplex) shows small but persistent deviations (JSD ≈0.0075–0.0078); MC–D α = 0.5 yields near-identity (JSD ≈0.0010–0.0011). Runtime trade-offs are explicit: FRWSA at h= 0.05 takes 105 s (and 317 s at h = 0.04); MC/MC–D reach 10^7 draws in 56–62 s. FRWSA is variance-free on the grid, whereas MC results are conditional on the sampling design.

FRWSA provides compact dominance-frequency summaries (share of the weight simplex where each option is best), decision-boundary diagnostics that localize rank switches, and a declared-resolution, reproducible baseline against which sampling or visual tools can be calibrated. The framework is method-agnostic (TOPSIS here; VIKOR in the SI shows comparable robustness patterns). Recommended reporting includes the chosen h, computation time, dominance frequencies, and boundary visualizations. Limitations are the dependence on grid resolution and combinatorial growth with large n. Promising directions include adaptive/coarse-to-fine enumeration and stratified scans near detected boundaries, scalable parallelization, applications with larger n and other cores (PROMETHEE, WASPAS, TODIM), and links to multi-objective analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.dajour.2025.100642.

Data availability

Data will be made available on request.

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